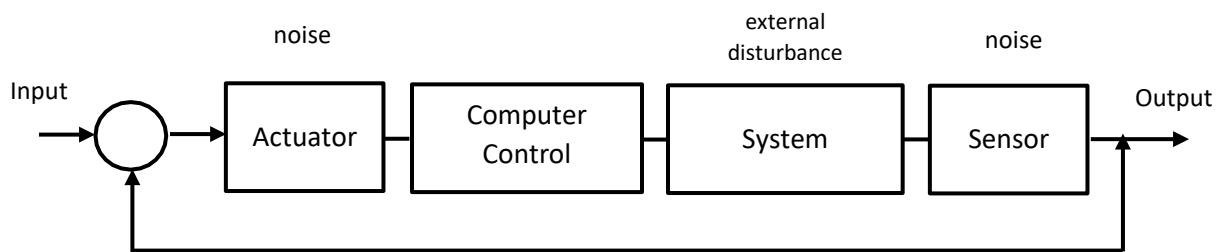


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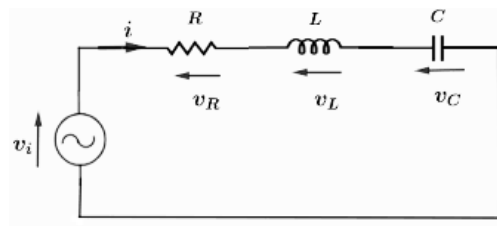
Part 2 : Tutorial 1 – Intro & System Modelling

Q1 : Identify any three feedback systems that you encounter in your everyday environment. For each system identify the sensing mechanism, actuation mechanism and control law. Describe the uncertainty with respect to which the feedback system provides robustness and/or dynamics that are changed using feedback.

Q2 : Consider balancing our self on one foot with our eyes closed for 15sec as a balance system. Using the block diagram below as a guide, describe the control system responsible for preventing you from falling.



Q3 : Derive the Transfer Function $\left(\frac{I(s)}{V_i(s)}\right)$ – in the Laplace Transform domain - of the RLC network as shown below :



RLC Network (2nd Order System)